Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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Abstract—**Artificial-intelligence (AI) data centers are flexible grid resources, but a meter-only demand-response baseline cannot show that credited work was committed before the event, nor can it distinguish a local reduction from spatial rebound. This paper develops an auditable submit–execute–settle verifier that carries one declared workload state from counterfactual scheduling to signed nodal settlement. A sparse cumulative linear program and an exact job-indexed witness enforce release, deadline, energy, graphics- processing-unit (GPU) nameplate, and regional-capacity constraints. The reported risk profile is a validation-fitted simplex under total and daily conditional-value-at-risk budgets; a separate submission-masked gate policy and a fixed/flexible power-conversion certificate keep execution telemetry out of contract formation. On the 121-day trace inventory, the locked mechanism-isolation panel gives normalized root-mean-square error (nRMSE) 0.314, false credit 1.809 MWh/day, and F_1 score 0.470, compared with 0.339, 1.823 MWh/day, and 0.418 for the selected single projection. The ledger contains 71,144 submissions and 71,128 matched executions; its indexed witness has 12,296,675 job–slot variables and a declared-energy residual below 10^{-15} MWh. A 54-day gate replay uses only pre-event declarations, while an independent network replay evaluates the same witness under all 37 finite RTS-24 contingencies and two homogeneous scale transforms. Fixed facility demand is carried separately, so the raw q_{99} endpoint remains an auditable activation scenario rather than a silently clipped stress case. The certificate reports workload feasibility, information provenance, and signed N–1 value without treating public traces as co-located utility measurements.**

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. Introduction

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch graphics-processing-unit (GPU) work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value. Public traces already expose multi-week bursty workload arrivals and tens of thousands of scheduler jobs, while facility telemetry reports GPU energy at a different measurement boundary [1], [2].

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice under published market guidance [3], despite known bias and manipulation incentives [4]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [4]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide the missing physical state. Temporal deferral, dynamic capacity, and geographical load balancing can reduce energy cost and peak demand. Recent work assesses data-driven flexibility from production traces and system-level security-constrained coordination, while balancing-market and clean-computing studies optimize spatial–temporal data-center dispatch [5], [6]. A recent power-system review also identifies quality-of-service (QoS), service-level-agreement (SLA), and market-participation constraints that are absent from meter-only accounting; clean-computing coordination provides a complementary dispatch context [7], [8]. Hierarchical optimal-consumption models further connect facility efficiency with response quality. Virtual-link market clearing and space–time remuneration establish how geographical and temporal transfers can enter electricity markets [9], [10]. Receding-horizon operation and uncertainty-aware frequency-regulation coordination further establish the need to retain future arrivals and multi-site state [11]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains unresolved. This separation between computing feasibility and market verification motivates the auditable ledger used here. Recent production-trace studies quantify data-center flexibility and load-aggregator coordination, but leave this ledger-to-settlement coupling open [12], [13]. In particular, cross-regional dispatchable-capacity evaluation and coupled-regulation modeling quantify operational flexibility but do not provide a provenance-bound settlement certificate [14], [15]. Recent whole-facility measurements likewise require a separate bottom-up conversion from GPU power to facility demand; the present study therefore keeps raw energy calibration, benchmark scaling, and network placement as separate evidence layers.

The second gap is electrical. Geographic workload control is commonly valued with energy prices or gross megawatts. Such accounting ignores the fact that nodal marginal value changes when congestion or a contingency constraint binds. Direct-current (DC) power flow and linear distribution factors provide transparent network sensitivities [16], [17]; N–1 dispatch can impose every credible post-contingency line limit through line-outage distribution factors (LODFs) [18]. No existing data-center DR verification framework jointly certifies workload feasibility, preserves signed spatial response, and settles the resulting security-aware network value. Alternating-current (AC) feasibility is a distinct nonlinear check, not a consequence of the DC certificate. Public workload and facility telemetry can support an observational replay of that ledger, but it does not constitute a causal utility event; a rigorous study must therefore separate measured-execution replay, counterfactual construction, and network-value verification to avoid conflating them with one field intervention.

This paper makes three power-system contributions. 1) It defines a typed submit–execute–settle evidence chain: an exact job-indexed counterfactual is formed only from scheduler declarations (release, runtime, requested GPUs, and declared energy entitlement), whereas the execution ledger is retained for post-event scoring and the settlement ledger is generated only after the coupling certificate passes. The indexed witness is independently rechecked before its regional profile is valued in the N–1 model. 2) It formulates a workload-state verifier with an explicit information boundary and a validation-fitted total-plus-CVaR risk fit, while keeping the pointwise false-credit cap in a separately auditable payment profile. A separately parameterized event policy and a binding-capacity stress cohort test whether the declaration-based feasible set remains nonempty when regional limits, service floors, and declared windows are jointly active. 3) It develops signed space–time nodal remuneration and a payment-activation rule over held-out power-conversion factors: the raw q_{99} endpoint is retained, but a payment can activate only when the precommitted fixed/flexible capacity certificate covers it. The central result is a dimension-preserving coupling invariant that carries one declared workload state into physical verification, network valuation, and settlement; the individual linear-program (LP), dispatch, and allocation primitives are standard and are not claimed as new in isolation. Because public traces do not identify utility geography, spatial conclusions are conditional on the declared feature- stratified trace scenario and the complete permutation/penetration panel. The certificate assumes trusted scheduler and telemetry provenance: a digest detects post-commitment record changes but cannot authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. System Model and Problem Formulation

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The measured ledger supplies arrival energy a_{skt} for source s, class k, and release interval t. The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ. Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^{t} \sum_{d\in\mathcal{D}} x_{skd\tau}. \tag{1}$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^{t} a_{sk\tau}$ and deadline D_k, release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \tag{2}$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \tag{3}$$

The state recursion equals service, terminal service equals total arrivals, and $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \tag{4}$$

For rolling window end T^+ and final event-day release T^0, the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk,\max\{T^0, T^+ - D_k\}}, \tag{5}$$

which completes every job due inside the window and every event-day arrival, while retaining later real arrivals in the release constraints. The window starts $D_{\max} + 96$ intervals before the event day and ends $D_{\max}$ intervals after it; omitted earlier work must therefore expire before the evaluated day. Equations (2) and (3) prevent early execution and late completion, while (4) maps the feasible service state to the facility meter. The aggregate variable x is a preemptive envelope for checkpointable batch work; contiguous job intervals are retained separately for the MIT replay and checked in Experiment 14. This two-level construction keeps the counterfactual optimization sparse while making the task-level scope explicit. For the task-level check, $j \in \mathcal{J}_{\text{sub}}$ has submitted release r_j, entitlement $\widehat{E}_j$, scenario region s_j, and requested GPU count g_j^{req}. Its endpoint is $d_j = r_j + R_j + B$, where $R_j = \lceil \text{timelimit}_j / \Delta t \rceil$, $B = 96$ precommitted queue slots, and the unlimited sentinel is mapped to $H_\infty = 128$ slots. Finite declarations are retained exactly; the 0.001-MW-per-GPU nameplate is an audit-only precheck, and observed completion never enlarges d_j. The indexed service variable $u_{j\tau}$ is constrained by

$$\begin{aligned}
&u_{j\tau} \geq 0, \qquad \sum_{\tau=r_j}^{d_j-1} u_{j\tau} = \widehat{E}_j, \\
&u_{j\tau} \leq g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t, \\
&\sum_{j: s_j = d,\ r_j \leq \tau < d_j} u_{j\tau} \leq \bar{P}_d \Delta t, \\
&\min_u \sum_{j,\tau} \left[w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\} \right] u_{j\tau}.
\end{aligned} \tag{6}$$

Here $\bar{p}_{\text{GPU}} = 0.001$ MW/GPU. Equation (6) is a globally solved, declaration-only schedule rather than a replay; its event tariff is fixed before scoring. Experiment 19 covers all 71,144 submissions, of which 71,128 have a positive-energy match used only for independent scoring. The job-to-network certificate maps this witness without a second optimization. With native assignment $d = s_j$, let $\mathcal{J}_{skd}$ be the corresponding ledger partition:

$$x^{\text{job}}_{skd\tau} = \sum_{j\in\mathcal{J}_{skd}:\, r_j\le\tau<d_j} u_{j\tau}, \tag{7}$$

$$p^{\text{job}}_{d\tau} = P^{\text{fix}}_d + \Delta t^{-1} X^{\text{job}}_{d\tau}, \qquad X^{\text{job}}_{d\tau} = \sum_{s,k} x^{\text{job}}_{skd\tau}. \tag{8}$$

Experiment 22 reconstructs X^{job} from every stored $u_{j\tau}$, rechecks the indexed primal, and verifies equality before network valuation; the aggregate migration profile is only a matched-workload control. Workload deferral and geographic assignment follow [5], [6]; rolling-state control follows [11]; waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$.

For provenance, the submit-time ledger and post-event execution ledger have separate canonical digests. Each submitted record is represented by the scheduler-only tuple and its digest in

$$\begin{aligned} \ell^{\text{sub}}_j &= (\text{id}_j, t^{\text{submit}}_j, \text{timelimit}_j, q_j, \kappa_j, \sigma_j),\\ \mathcal{L}_{\text{sub}} &= \text{sort}\{\ell^{\text{sub}}_j : j\in\mathcal{J}_{\text{sub}}\},\\ h_{\text{sub}} &= \text{SHA256}(\text{serialize}(\mathcal{L}_{\text{sub}}))\,. \end{aligned} \tag{9}$$

where q_j, κ_j, and σ_j are requested-resource, class, and scheduler-state fields. The execution ledger stores start/end times, DCGM energy, and counts; its join and energy equality are checked only post-event. Canonical ordering and the Secure Hash Standard [19] keep h_{sub} in (9) independent of execution telemetry. Experiment 16 reconciles the measured envelope with the pre-split 118-MW nameplate. Scheduler/DCGM records are authoritative; the digest detects changes but cannot attest semantic truth of submitted fields.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p^0 is the no-event counterfactual, $p^{1,\text{sim}}$ the workload event trajectory, and p^{obs} the independent MIT/DCGM replay. Because no utility event label is available, the replay is observational only and never a causal outcome or pre-decision input. In mechanism-isolation rows, signed rebound is retained as $r = p^0 - p^{1,\text{sim}}$. For source z in the observed or simulated replay, define $\hat{r}^{(z)} = [\hat{p}^0 - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; source-specific false credit is

$$F^{(z)} = \Delta t \sum_{d,t\in\mathcal{E}} [\hat{r}^{(z)}_{dt} - r^{(z)}_{dt}]_+, \qquad z\in\{\text{obs},\text{sim}\}. \tag{10}$$

The simulated source defines the validation epigraph; the observed source is recomputed only in locked replay. The identity $p^{(\text{sim})} + r^{(\text{sim})} = \max\{p^0, p^{(\text{sim})}\}$ is audited but is not a new meter trajectory. Before the event the operator freezes p^{con}; after closure, payable response is

$$Q^{\text{pay}} = \Delta t \sum_{d,t\in\mathcal{E}} \min\bigl\{[\hat{p}^0_{dt} - p^{\text{obs}}_{dt}]_+, [p^{\text{con}}_{dt} - p^{\text{obs}}_{dt}]_+\bigr\}\,. \tag{11}$$

Equation (11) uses only the submitted baseline, frozen contract, and closed meter. The trace-anchored p^0 is an offline diagnostic and never enters fitting, gate decisions, workload optimization, or transfer formation. Reported metrics include gross/payable response, oracle over/underpayment, nRMSE, bias, precision, recall, and F_1.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q, and DR price π^{DR}. If reference day j adds event-window service δR_j, the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\text{DR}}}{H}\sum_{j=1}^{H}\delta R_j - \sum_{j=1}^{H}\Delta C_j. \tag{12}$$

Let c'_j be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day j. Linear-program sensitivity [20] gives the following exact local condition:

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \tag{13}$$

At a degenerate optimum the dual at the current service level can be zero even when the right-direction marginal cost is positive. Experiment 1 therefore certifies c'_j with two independent right finite differences of 5×10^{-4} and 10^{-3} MWh and a 10^{-6} USD/MWh agreement tolerance. The historical-baseline premise is documented in [4]; (12)–(13) are this paper's workload-specific specialization.

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p^{\text{native}}_t + Mp^{\text{dc}}_t$ the nodal demand vector. For generator-to-bus map C_g, generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \ & \sum_i C_i(g_i) \\ \text{s.t. } & \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g}\le g\le\bar{g}, \\ & -\bar{f}\le H(C_g g - p)\le\bar{f}. \end{aligned} \tag{14}$$

The incidence mapping aggregates device demand at its connection bus, and (14) is the standard lossless DC representation [16], [17]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \tag{15}$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. Proposed Methodology

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations in Section II are the first stage of the methodology: they define the feasible workload state before any statistical target or payment is formed. Fig. 1 shows the complete chain. Meter history and the submitted-job ledger enter a statistical pre-estimator and the workload feasible set in parallel. The pre-estimator never receives execution energy used as scoring truth. For each event day, it uses only the information available under one of two declared protocols: lagged meter/calendar features for online baselines, or the complete submitted ledger for equally informed post-event comparisons. Ridge regression and gradient boosting are used as standard matched-information comparators [21], [22]; extremely randomized trees and synthetic control provide two additional nonlinear and donor-pool comparators [23], [24]. Two stronger comparators are also solved: median-loss gradient boosting with the complete submitted ledger and a nonnegative synthetic control whose weights globally minimize pre-event absolute error over 28 donor days. To separate the effect of projection from that of pre-estimation, the complete-ledger quantile output is also passed through the same exact workload-feasible projection, with its penalty chosen by contiguous validation folds. The same scheduler declaration view is supplied to the post-event learner, the selected single projection, and the proposed convex verifier; execution energy and completion timestamps are withheld until the locked scoring pass.

B. Sparse cumulative workload-state verifier

For statistical estimate $\widetilde{p}$ and each predeclared penalty ρ_ℓ, the first-stage program solves

$$\min_{x,y,z}\ \mathcal{C}_{\mathrm{w}}(x)+\rho_\ell\sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt}\ge \pm(p_{dt}(x)-\widetilde{p}_{dt}), \tag{16}$$

subject to (1)–(4). The absolute-value epigraph is standard [20]; its use to project a meter baseline onto the auditable workload set is the study-specific step. Six globally solved schedules $p^{(\ell)}$ share all arrivals and constraints, so their validation-only simplex combination $\bar{p}=\sum_\ell \alpha_\ell p^{(\ell)}$ remains feasible.

The coefficients minimize validation squared error subject to total and daily-tail mechanism-isolation false-credit budgets; conditional value at risk (CVaR) measures the latter [25]. Let $\ell^\star$ be the single candidate with minimum contiguous-fold nRMSE and $B_j^{\mathrm{cap}}=F_j^{(\mathrm{sim})}(\ell^\star)$. The observed-meter source is audited only after the locked decision. For reserve $\beta\in(0,1]$,

$$\sum_j F_j^{(\mathrm{sim})}(\alpha)\le \beta\sum_j B_j^{\mathrm{cap}}, \tag{17}$$

$$\mathrm{CVaR}_{0.75}\Big(F_j^{(\mathrm{sim})}(\alpha)\Big)\le \beta\mathrm{CVaR}_{0.75}(B_j^{\mathrm{cap}}). \tag{18}$$

Implementation linearizes the positive part with $f_n\ge 0$, a CVaR threshold $\nu\ge 0$, and tail slacks $u_j\ge 0$ for each event sample and validation day:

$$f_n\ge p_n^{\alpha}-p_n^{(\mathrm{sim})}-r_n^{(\mathrm{sim})}, \tag{19}$$

$$F_j^{(\mathrm{sim})}=\sum_{n\in j} f_n, \tag{20}$$

$$F_j-\nu-u_j\le 0, \tag{21}$$

$$\nu+\frac{1}{0.25J}\sum_j u_j\le \beta\,\mathrm{CVaR}_{0.75}(B_j^{\mathrm{cap}}). \tag{22}$$

Here $r_n^{(\mathrm{sim})}=[p_n^0-p_n^{(\mathrm{sim})}]_+$; the explicit simulated and true terms remain in the model. The epigraph is never substituted for the frozen contract profile p^{con}. Together with the simplex and box constraints, (22) is a convex quadratic program. HiGHS supplies a feasible point, after which stationarity, epigraph, simplex, and bound residuals are recomputed and stored. Four contiguous folds select β by validation nRMSE and then risk ratio; held-out noninferiority is an outcome, not a hidden constraint. A separate payment-contract projection targets $\bar{p}$ while imposing, for every event sample,

$$p_{dt}^{\mathrm{safe}}\le p_{dt}^{(\ell^\star)}. \tag{23}$$

The pointwise cap in (23) is the contractual upper band, evaluated on every event slot before settlement and never substituted for the risk profile.

a) Two-sided credit band.: The upper envelope is paired with a lower band tied to the independently fitted target. Define $p_{dt}^{\mathrm{floor}}=\min\{p_{dt}^{(\ell^\star)},\bar{p}_{dt}\}$; the event-slot constraints are

$$p_{dt}^{\mathrm{floor}}-\varepsilon\le p_{dt}^{\mathrm{safe}}\le p_{dt}^{(\ell^\star)}, \qquad (d,t)\in\mathcal{D}\times\mathcal{E}, \tag{24}$$

where $\varepsilon=1$ MW is fixed before the locked split and imposed with the workload constraints, not clipped afterward. The total and CVaR constraints control aggregate and daily-tail exposure; the upper side is the deterministic payment cap, while the lower side is an under-credit regularizer rather than a distribution-free unseen-trace guarantee. The selected single projection proves nonemptiness, and locked outputs report both margins and payable response.

b) Decision-time information boundary.: The ex-post selector may use the complete ledger, but a gate deployment removes all post-gate arrivals before target construction. The masked target is refit with the matched-information model; a no-tariff LP freezes the baseline and commits 5 MWh, while a second LP applies the DR price with unexpired state. Reserved future jobs cannot be credited. Baseline error is scored only for the submitted contract; the tariff plan is scored against the closed meter and settled by (11).

To separate capacity planning from payment eligibility, the gate publishes a causal reserve envelope. For candidate quantile η and event day d,

$$\widehat{a}_{sk\tau}^{\mathrm{res}}(d;\eta)=Q_\eta(\{a_{sk\tau}(h):h<d\})\,, \qquad \tau\ge t_{\mathrm{gate}}, \tag{25}$$

Equation (25) uses earlier days only. Candidates are validation-selected for capacity planning; no reserved job enters payment before ledger commitment.

The cumulative state reduces explicit-prefix nonzeros from quadratic to $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$. Equation (5) retains observed future arrivals in the 512-slot continuous-horizon certificate. A lexicographic pair of linear programs first minimizes L1 distance and then operating cost on that face, so no penalty weight trades fit against cost; all programs are solved globally with HiGHS [26]. Fig. 1 summarizes the chain and Fig. 2 details the ledger-to-LP path; execution replay and interval audits remain separate experiments.

C. Signed nodal and N–1 settlement

Let $\lambda^0\in\partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top(p^0-p^1)$; (15) is the realized system-value benchmark and bilateral contract payoff, not an ISO market payment. The convex subgradient inequality [20] yields

$$0\le(\lambda^0)^\top(p^0-p^1)-\Delta V=V(p^1)-V(p^0)-(\lambda^0)^\top(p^1-p^0). \tag{26}$$

Thus the linear rule upper-bounds exact value in the implemented polyhedral model; equality requires a common supporting dual subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from anticipatory shift through recovery. The space–time rule is

$$\Pi^{\mathrm{ST}}=\Delta t\sum_{t\in\mathcal{W}}\sum_d \lambda_{dt}^0(p_{dt}^0-p_{dt}^1). \tag{27}$$

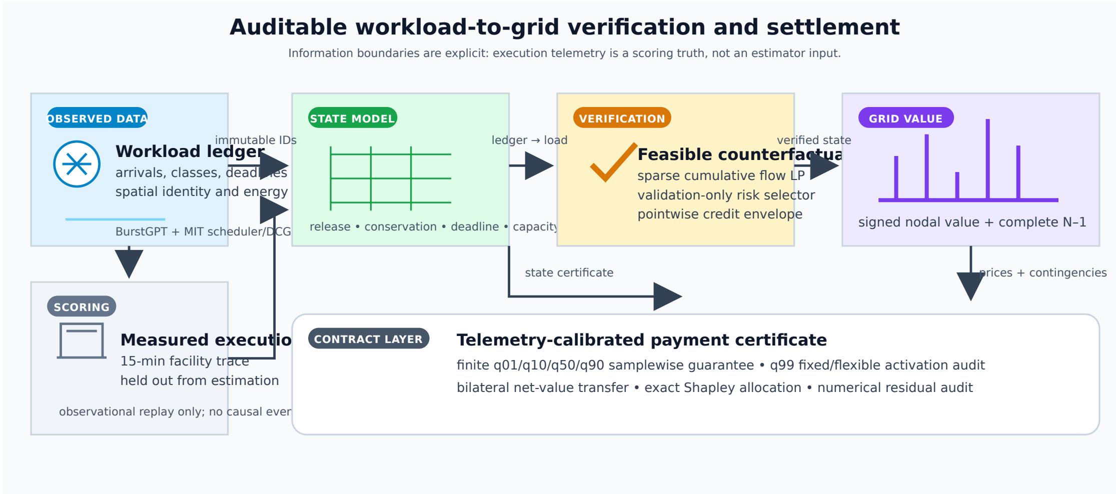

Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N–1 value and exact participant shares.

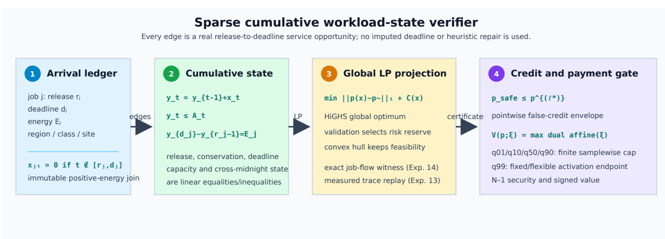

Fig. 2. Sparse cumulative verifier detail. Each admissible edge is a declared release-to-deadline service opportunity; the global projection, risk fit, and payment gate operate on the same feasible state and do not repair missing deadlines or execution records.

This applies nodal virtual-link remuneration [9], [10] to the verified counterfactual; summing site payments reproduces (27).

Space–time energy remuneration is one component of the contracted DR product. For participating sites $\mathcal{D}^{\text{part}}$, event intervals $\mathcal{E}$, and cleared price r^{DR}, delivered capacity and operator value are

$$Q^{\text{cap}} = \Delta t \sum_{d\in\mathcal{D}^{\text{part}}} \sum_{t\in\mathcal{E}} (p^{\emptyset}_{dt} - p^{1}_{dt}), \tag{28}$$

$$G = r^{\text{DR}} Q^{\text{cap}} + \Pi^{\text{ST}}, \tag{29}$$

where $p^{\emptyset}$ is the minimum-cost no-event schedule under the same arrival window. Capacity credits and ex-post participation follow established DR mechanism design [27], [28]; the signed Π^{ST} term debits transfer and recovery. Equations (28)–(29) define the operator value.

Let x^1 and $x^{\emptyset}$ induce p^1 and $p^{\emptyset}$; participant opportunity cost is $C_{\text{opp}} = \mathcal{C}_{\text{w}}(x^1) - \mathcal{C}_{\text{w}}(x^{\emptyset})$ and surplus is $S = G - C_{\text{opp}}$. With zero disagreement utilities, the symmetric Nash solution [29] activates iff $S \geq 0$ and sets

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\text{B}} = z\left(C_{\text{opp}} + \frac{S}{2}\right), \quad u_{\text{dc}} = u_{\text{op}} = z\frac{S}{2}. \tag{30}$$

Thus (30) makes nonactivation explicit; activated contracts have nonnegative utilities and an exactly balanced transfer, reported separately from ISO nodal remuneration and exact dispatch value. The complete window $\mathcal{W}$ also debits pre-event shifts and rebound, with $\sum_{d,t\in\mathcal{W}}(p^0_{dt} - p^1_{dt})\Delta t = 0$ up to tolerance.

Security-aware settlement augments (14) with every finite non-islanding outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ, then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k} H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \tag{31}$$

Equation (31) is the standard preventive N–1 LODF formulation [18]; all credible rows are imposed simultaneously. Islanding outages are reported but excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the DC planning criterion [30]; native ratings are retained.

D. *Scenario-robust N–1 payment certificate*

The pointwise envelope controls credited energy but not network payment when different demand vectors activate different security constraints. Let $\mathcal{P} = \text{conv}\{p^{(1)}, \ldots, p^{(L)}\}$ contain the workload-feasible schedules and let p^{cap} be the contiguous-validation single projection. The feasible-quantile profile remains an external transfer comparator. Let Ξ_4 be the held-out 1st/10th/50th/90th percentiles of measured-to-predicted GPU energy; these scenarios define the telemetry-backed conversion contract [2]. For facility profile p^{fac}_{dt}, fixed demand P^{fix}_d, and benchmark scale s^{dc}, the nodal injection under factor ξ is

$$p^{\text{dc},\xi}_{dt} = s_{\text{dc}}\left(P^{\text{fix}}_d + \xi[p^{\text{fac}}_{dt} - P^{\text{fix}}_d]\right), \qquad p^{\text{fac}}_{dt} \geq P^{\text{fix}}_d. \tag{32}$$

The scale s_{dc} is calibrated on the held-out q_{99} flexible peak; the fixed component is invariant to ξ, and raw q_{99} is retained without clipping. The flexible-only factor in Experiment 16 is diagnostic. Below, $Mp^{\text{dc},\xi}_t$ is the nodal injection; the cap profile and simplex profile use (32). For each day, payment-certified verifier defines $B^{\text{cap}}_\xi = \Delta t \sum_{t\in\mathcal{E}} V_{N-1}(p^{\text{native}}_t + Mp^{\text{dc},\xi,\text{cap}}_t)$

and solves

$$\operatorname{lexmin}_{\alpha,e,\eta,\{g_t^\xi\}} \quad \left(\sum_{d,t\in\mathcal{E}} e_{dt}, -\eta\right) \tag{33}$$

$$\begin{aligned}
\text{s.t.}\quad & p^\alpha = \sum_{\ell=1}^{L} \alpha_\ell p^{(\ell)}, \\
& \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \leq \eta \leq 1, \\
& e_{dt} \geq \pm(p_{dt}^\alpha - p_{dt}^{\text{safe}}), \\
& g_t^\xi \text{ satisfies (14) and (31),} \\
& \qquad \text{with } p_t = p_t^{\text{native}} + M p_t^{\text{dc},\xi,\alpha}, \\
& \qquad\qquad t \in \mathcal{E}, \quad \xi \in \Xi_4, \\
& \Delta t \sum_{t\in\mathcal{E}} \sum_i C_i(g_{it}^\xi) \\
& \qquad \leq (1-\eta) B_\xi^{\text{cap}}, \quad \xi \in \Xi_4.
\end{aligned}$$

The same piecewise-linear epigraph is used for settlement. Two linear programs first minimize trajectory error and then maximize the common fractional margin η on that face. Feasible dispatches in (33) give the upper bound on optimal N–1 value; unit weight on p^{cap} proves nonemptiness. Thus, for any $\xi \in \Xi_4$ and realized load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned}
&\sum_{t\in\mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M p_t^{\text{dc},\xi,\alpha}) - V_{N-1}(p_t^{1,\xi})] \\
&\leq \sum_{t\in\mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M p_t^{\text{dc},\xi,\text{cap}}) - V_{N-1}(p_t^{1,\xi})].
\end{aligned} \tag{34}$$

The realized-cost term cancels, so the certificate needs no execution truth and holds samplewise; every finite non-islanding contingency is included. Standard SCED and epigraph foundations are used [17], [20]. $\mathcal{P}$ contains the six first-stage projections and the feasible-quantile projection. risk-constrained verifier is the validation-fitted simplex point under total and CVaR budgets; the separately stored payment-contract profile uses the pointwise cap and is not fed back into the risk fit.

The finite cap is complemented by a set-valued payment certificate. Let $p^{\text{cap}} = p^{(\ell^\star)}$ be the contiguous-validation cap and let p^{pay} be the independently payment-selected target. For $\Delta V_\xi(p) = V_{N-1}(p,\xi) - V_{N-1}(p^{1,\xi})$ and $p_\theta = (1-\theta)p^{\text{cap}} + \theta p^{\text{cert}}$, the induced interval is

$$\mathcal{I}_\xi = \left[\min_{0\leq\theta\leq1} \Delta V_\xi(p_\theta),\ \max\{\Delta V_\xi(p^{\text{cap}}), \Delta V_\xi(p^{\text{cert}})\}\right]. \tag{35}$$

The lower endpoint is one exact joint N–1 LP with shared segment parameter; convexity gives the upper endpoint. The cap payment lies in $\mathcal{I}_\xi$. Endpoints use validation-frozen profiles and the realized counterfactual only; no oracle baseline selects or certifies the interval. Its seven-profile hull is the finite-scenario certificate in (33).

E. Participant allocation

For participants N, let $v(S)$ be signed dispatch value when coalition S moves from baseline to actual demand. The exact allocation is

$$\phi_i(v) = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!}[v(S\cup\{i\}) - v(S)]. \tag{36}$$

This is the Shapley value [31]; signed grid value is the study-specific characteristic function. Grouping (36) by coalition size gives $\sum_i \phi_i = v(N)$ for $v(\varnothing) = 0$. All 16 four-site and 256 eight-site coalitions are enumerated exactly. Larger contracts use exchangeable site slices with binomial multiplicity, evaluating only $\prod_g (m_g+1)$ states; the exact value is checked through 20 participants.

F. Analytical guarantees

The certificates make the scope explicit. Under the assumptions of (12), a reference-day perturbation is profitable iff $q\pi^{\text{DR}}/H > \min_j c'_j$ in (13); Experiment 1 checks this against the independently solved 56-cell grid. The workload set is convex, so every simplex schedule and (24) are feasible. For any execution c,

$$\begin{aligned}
[p_{dt}^{\text{safe}} - c_{dt}]_+ &\leq [p_{dt}^{(\ell^\star)} - c_{dt}]_+, \\
[c_{dt} - p_{dt}^{\text{safe}}]_+ &\leq [c_{dt} - p_{dt}^{\text{floor}}]_+ + \varepsilon.
\end{aligned}$$

Summation gives both bounds without locked outcomes.

The base-case and stacked N–1 values are convex piecewise-affine functions of nodal demand. Equation (26) gives $0 \leq \Delta V_{\text{lin}} - \Delta V \leq \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|$; positive-only settlement has no such identity. In (33), each embedded dispatch is a feasible witness, so subtracting the common realized cost gives (34) for every $\xi \in \Xi_4$. The dual representation also makes $V_{N-1}(p;\xi)$ convex in a continuous conversion factor, so the worst case on $[\xi_{\text{L}}, \xi_{\text{U}}]$ is attained at an endpoint. The lower endpoint of (35) is the exact joint-LP minimum over the declared affine segment; convexity makes its endpoint maximum exact. Experiments 9 and 15 use independent 40- and four-segment models.

Finally, SHA-256 makes post-commit changes to the scheduler tuple detectable in h_{sub}, while joins check ordering and energy conservation after the event. An optimum of (6) enforces entitlement, submit-time windows, site capacity, and the requested-GPU nameplate without aggregate rounding; contiguous execution is audited separately. With $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map M, the job, aggregation, and mapping residuals are

$$\begin{aligned}
r_j^{\text{job}} &= \sum_{\tau=r_j}^{d_j-1} u_{j\tau} - \widehat{E}_j, \\
r_{d\tau}^{\text{agg}} &= X_{d\tau} - \sum_j B_{dj} u_{j\tau}, \\
r_{b\tau}^{\text{map}} &= p_{b\tau} - P_{b\tau}^{\text{fix}} - \Delta t^{-1} \sum_d M_{bd} X_{d\tau}.
\end{aligned}$$

If these typed norms are at most ε, the profile is the same indexed witness up to tolerance; at zero residual an independently optimized aggregate cannot receive settlement value. Experiment 22 checks the residuals before N–1 valuation and preserves $\sum_i \phi_i = v(N)$.

IV. Experimental Design

A. Data, preprocessing, and information boundary

BurstGPT supplies 5,188,507 successful requests and token service [1]; MIT Supercloud supplies scheduler submissions and independent Data Center GPU Manager (DCGM) energy. Deterministic validation leaves 71,144 submissions and 71,128 positive-energy matches, with 68,664 starting in the common 121-day tensor [2]. Submission time defines arrivals and measured intervals define scoring truth. A held-out GPU-power model (28,413 observations) gives $R^2 = 0.899$, MAE 11.22 W, and RMSE 15.82 W; scaling uses the first 40 complete training days. Held-out ratios define Ξ_4 in (33); Experiment 14 retains all matches, whereas Experiment 19 uses declarations and a training-only entitlement. The execution join is opened only for post-event scoring. Its runtime window plus the precommitted 96-slot queue allowance is a contract-window definition, not a submit-to-completion interpretation of Slurm "timelimit".

Because utility bus locations are unavailable, four-region coupling is a trace-driven benchmark: a feature-stratified round robin assigns class, GPU count, runtime, and submission time. Experiment 11 exhausts 24 permutations, three penetrations, and two concentration controls; Experiment 18 uses fixed generator-bus role strata at native-load multiplier 0.90 and is therefore conditional on the declared trace-to-bus scenario. Held-out ratios span 0.5845–2.6931. Equation (32) carries the fixed 6-MW/site component separately and calibrates on q_{99}, leaving both raw endpoints activation-eligible; the 0.8901 flexible-only factor is planning diagnostic. PGLib/MATPOWER/PYPOWER provide topology, ratings, and costs [17], [32]; RTS-24 follows the published case [33].

B. Baselines, parameters, and inference

Experiment 2 compares eight meter/statistical baselines with complete-ledger quantile, feasible, synthetic-control, tail-risk counterfactual, single-projection, and risk-constrained verifier profiles; payment-certified verifier is evaluated separately in Experiments 9 and 15 because its hull and N–1 objective differ. Feature equality is audited with execution truth withheld. Deferral windows are 0, 4, and 512 slots; each region has a 118-MW flexible nameplate and 6-MW fixed component. Six projection penalties, four risk reserves, and four response weights are validation-only. The gate commits 5 MWh, payment uses an independent 40-segment evaluator, and three-day bootstrap/sign tests use Holm correction [34], [35]. Sparse-system size grows from 5,760 variables/13,424 nonzeros at 96 slots to 36,480/85,488 at 608 slots (0.039–0.212 s).

The model-out check uses polar alternating-current optimal power flow (AC OPF) with active/reactive, voltage, and apparent-power limits [17]. After the intact solve, every outage imposes for each non-reference generator i,

$$P_{Gi}^{(k)} = P_{Gi}^{(0)}, \qquad i \notin \mathcal{G}_{\text{ref}},\ k \in \mathcal{C}. \tag{37}$$

The reference generator balances active losses; $Q_G^{(k)}$ and voltages remain recourse. Limits are checked after each fixed-plan solve, so (37) tests one shared active plan. Non-reference bounds are fixed at intact output $\pm 10^{-8}$ MW with a 10^{-6}-MW residual margin; the largest deviation is 2.52×10^{-8} MW.

Experiments 1–2 test incentives and workload verification; 3–12 test signed settlement, transfer, risk, N–1/AC value, and continuous-cycle remuneration; 13–16 provide measured replay, indexed witness, conversion intervals, and ledger reconciliation. Experiments 17–25 freeze the gate, verify AC and job/network coupling, audit scale, replay an independent event, and enumerate RTS-24 outages. Confirmatory comparisons use the same 54 locked days unless a deterministic network or ledger rule is stated.

TABLE I
LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS).

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	4.196	172.455	6.474	0.143
Ridge	2.761	109.817	5.306	0.233
Gradient Boosting	2.343	96.077	4.817	0.245
Extra Trees	3.350	133.961	6.804	0.171
Metadata Gradient Boosting	2.201	92.874	4.193	0.259
Ex-post Metadata Gradient Boosting	2.126	90.807	4.184	0.265
Ex-post Quantile Gradient Boosting	0.932	23.144	6.085	0.533
Synthetic Control	2.598	104.998	7.877	0.202
Feasible Quantile Projection	0.301	2.811	10.772	0.519
Single Feasible Projection	0.339	1.823	13.537	0.418
Tail-Risk Feasible Counterfactual	0.296	2.521	10.864	0.518
Risk-Constrained Convex Verifier	**0.314**	**1.809**	**12.517**	**0.470**

V. RESULTS

A. Incentive boundary and independent verification

Experiment 1 solves the complete 8×7 DR-price/call-probability grid. The dual threshold (13) matches the strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, reference scheduling inflates the event baseline by 8.223 MWh; 44.458 of 55.452 MWh credited is unsupported (ratio 0.802). The profitable region follows the derived marginal boundary without a fitted classifier.

Experiment 2 evaluates twelve methods on 54 later days (Table I). The CVaR-only ablation changes (nRMSE, F_1, false credit) from $(0.339, 0.418, 1.823)$ to $(0.296, 0.518, 2.521)$ MWh/day, while the joint total-plus-CVaR fit gives $(0.314, 0.470, 1.809)$. The latter is risk-constrained verifier; its payment cap is stored separately. Against feasible quantile, false credit falls 2.811 to 1.809 MWh/day while nRMSE rises 0.301 to 0.314, exposing the risk–accuracy frontier. Relative to the single reference, total-plus-CVaR changes (nRMSE, false credit, F_1) by $(-0.025, -0.014, +0.051)$ and CVaR-only by $(-0.043, +0.698, +0.100)$; removing both constraints gives 3.241 MWh/day.

The independent trace-meter replay is a deployment check: event-window MAE is 2.673/2.651/2.810/2.988 MW for feasible-quantile, tail-risk counterfactual, risk-constrained verifier, and single-projection profiles on the same 54 days. It measures closed-DCGM alignment, not the mechanism-isolation panel.

Closest domain baselines use identical arrivals, deadlines, capacities, event slots, and locked days. The incentive-compatible spatial-DR analogue gives $(\text{nRMSE}, \text{false credit}, F_1) = (3.068, 114.776, 0.137)$; the flexibility/receding-horizon controls give $(0.339, 1.823, 0.418)$. Han-style joint-capacity and Chen-style coupled-regulation analogues are structural controls (the latter and the all-site exact comparator give $(0.326, 0, 0.285)$), not software reimplementations [14], [15]; receding-horizon translation follows [11]. Temporal-only and joint spatio-temporal controls have migration 0 and 0.206 MWh. The ablation panel contains 810 global schedules (binding shares 0.310/0.948; slope 1.003).

TABLE II
BASE-CASE AND COMPLETE N–1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Nodal gross	360.580	421.677	–	–
Base case	Nodal exact net value	**151.719**	455.865	–	–
N–1	Base exact	139.221	167.367	79.79/130.42	121.01/107.69
N–1	N–1 linear	2.208	**150.461**	0.60/1.91	113.43/20.38
N–1	**N–1 exact**	**0.990**	150.955	0.87/0.01	113.43/20.18

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms (Table II). Exact signed nodal value has MAE \$151.719 versus \$360.580 for gross response; workload errors are \$455.865 and \$421.677. Signed netting reduces trace error by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$), while uniform-to-nodal and linear-to-exact changes are $-\$1.463$ ($p = 0.0118$) and \$0.671 ($p = 0.8026$); workload effects are $-\$34.165$, $-\$1.191$, and $-\$0.023/day$. All 5,616 instances satisfy (26), with maximum violation 2.03×10^{-11} dollars.

Experiment 4 applies the predeclared largest gross-minus-net offset (day 87), exposing the rebound component that positive-only local accounting omits.

C. Cross-network and N–1 security evidence

Experiment 5 evaluates 6,480 four-network outcomes; exact value improves all 12 trace cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05) with a \$0.050/day workload change. Experiment 6 solves 810 capacity/deadline schedules (binding shares 0.031–0.911; false credit 2.374–2.740 MWh). Experiment 7 enumerates 16 four-site, 256 eight-site coalitions, and 1,296 exact 20-participant states (maximum residual 5.68×10^{-14} dollars). Experiment 8 imposes (31) for all 37 RTS-24 non-islanding outages: trace/workload MAE is \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 for base exact/N–1 linear/N–1 exact, with 1.000 p.u. maximum loading.

Experiment 9 tests (34) with 54 global programs and a validation-frozen seven-profile hull. Contiguous-fold baseline nRMSE selects the cap, while independent validation N–1 payment MAE selects the target; locked outcomes are unopened. Held-out factors 0.584/0.734/0.996/1.278/2.693 have cap violations below 3×10^{-9} USD. Below q99, independent 40-segment MAE is \$70.792/\$59.309/\$62.069/\$60.073 per day for payment-certified verifier, single, feasible-quantile, and risk-constrained verifier; mean overpayments are \$2.947/\$7.727/\$6.954/\$7.502. At q99, MAE is \$214.314/\$179.816/\$188.315/\$182.297 for the same methods, with the cap proof unchanged. Experiment 15 retains raw $q_{01} = 0.5845$ and $q_{99} = 2.6931$; fixed/flexible conversion (32) makes both endpoints activation-eligible under q99 calibration without clipping. The flexible-only 0.8901 factor is diagnostic. Unseen 0.80/1.20 transfer gives certified MAE \$16.170/\$24.262 and feasible-quantile MAE \$17.108/\$25.678. The conversion, preventive-AC, and scale panels use the same resource semantics in Table III.

Table III separates the preventive DC certificate, the shared-active-plan AC envelope, and the all-outage AC replay.

Figure 3 visualizes the same three evidence layers.

Experiment 10 replaces the linear network with AC OPF [17]: 864 base and 5,400 corrective IEEE-9/14 outage outcomes satisfy the limits. Experiment 18 separately fixes the intact active plan; its 3,888 cells have zero active deviation and maximum loading 0.880/0.907/0.935 p.u. at 3%/6%/9%. Experiment 11 spans 24 permutations, two concentration controls, three penetrations, four counterfactuals, and 54 days; averaged over 26 assignments, risk-constrained verifier MAE is \$98.839/\$198.726/\$296.240 at 3%/6%/9%, versus \$105.250/\$211.555/\$315.490 for single and \$81.613/\$164.249/\$244.481 for feasible quantile. Experiment 16 audits 66,769/28,413 training/test jobs (MAE 11.223 W, RMSE 15.818 W, $R^2 = 0.8994$). Experiment 12 reports event-window MAE \$26.937/\$15.969/\$5.325/\$5.325 and complete-cycle residual \$0.621/\$0.607/\$0.000/\$0.000 for feasible-quantile, payment-certified, RiskSafe, and single, respectively; the zero residual is an accounting certificate on the independent optimum face. All 54 contracts activate.

TABLE III
CROSS-NETWORK SECURITY PANELS AND DECLARED RECOURSE SCOPE.

Panel	Evaluated cells	Maximum loading / status
Preventive DC N–1 (RTS-24)	37 outages; 864 cells	1.0000 p.u.; all feasible
Shared-active-plan AC (IEEE-9)	3,888 cells	0.880/0.907/0.935 p.u. at 3/6/9%
Corrective AC replay (IEEE-9/14)	864 base + 5,400 outage cells	all connected cells feasible
All-outage AC replay (RTS-24/30/39/118)	864 cells; 31,968 solves	1.0000 p.u.; all feasible

TABLE IV
INFORMATION-BOUNDARY AND LOCKED RESPONSE CERTIFICATES (54-DAY MEANS). THE INDEPENDENT POLICIES USE A SUBMISSION-MASKED LEDGER, DISTINCT TARIFFS, AND NO VERIFIER OUTPUT; THE ROWS THEREFORE TEST CONTRACT/RESPONSE SEPARATION RATHER THAN A FIELD TREATMENT EFFECT.

Method	nRMSE	Gross false MWh	Payable MWh	Meter underpay	Recall	F1
Independent gate-committed response	2.606	145.506	227.116	94.055	0.436	0.358
Independent risk-constrained response	1.710	29.952	38.378	167.240	0.038	0.054
Complete-ledger risk-constrained verifier (mechanism isolation)	0.339	1.823	10.009	13.537	0.335	0.418

D. Independent trace and deployment-boundary audits

Immutable MIT scheduler/DCGM replay gives slot MAE 3.740 versus 5.678 MW at 100% coverage and a 71,128-job residual below 10^{-17} MWh. Observational trace-meter MAE is 2.988 MW for risk-constrained verifier (2.673 for feasible quantile); no utility event label is available. Experiment 23 applies a predeclared tariff and independent LP to a submission-masked ledger with no target, risk envelope, hull, or verifier output. Across 54 days the gate-committed response gives nRMSE/false credit/payable/precision/recall/F_1 =2.606/145.506/227.116/0.351/0.436/0.358, while the independent policy gives 1.710/29.952/38.378/0.188/0.038/0.054 (Table IV). This is a structural non-reuse probe, not a superiority or field-treatment comparison. Validation selects \$10/MWh, weight 1.0, and a 20-MWh/day budget; reserve $\eta = 0.60$ is planning-only. Experiment 18 supplies 9,648 fixed-plan AC-OPF outcomes at multiplier 0.90 with zero voltage violation and maximum loading 1.000000 p.u.

E. Indexed workload counterfactual

Experiment 19 assigns each submitted job and admissible slot a variable. Its submit-time runtime-plus-96-slot window, 0.001-MW/GPU bound, declared region, entitlement, and 118-MW

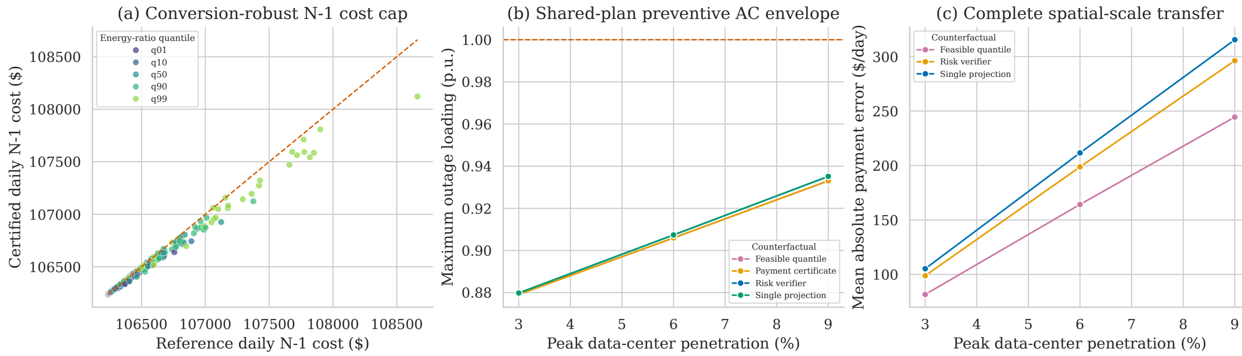

Fig. 3. Cross-layer robustness evidence. (a) The four-factor payment certificate remains below its reference cost. (b) The shared-active-plan AC envelope stays below the unit loading bound at all declared penetrations. (c) Spatial transfer error is reported across the complete 24-mapping panel and three penetrations.

TABLE V
EXACT INDEXED JOB-LEVEL COUNTERFACTUAL AND BINDING-CAPACITY CERTIFICATES.

Quantity	Value
Submitted / execution-matched jobs; job–slot variables	71,144 / 71,128; 12,296,675
Native / counterfactual event energy; gross/net; rebound (MWh)	0.194 / 0; **0.194/0.194**; 0
Maximum job residual; minimum site slack (MWh)	0; 29.486
Binding stress cohort / capacity / service floor (jobs/MW/MWh)	689 / 0.001 / 0.004
Capacity-proportional / fixed-nameplate scale (×)	2146.204 / 1.480
Job-to-network residual; coupled event slots (MWh / count)	0; 1,056
Secure N–1 value, raw / fixed / capacity scale (USD/interval)	**0.004293** / 0.006352 / 9.213788
RTS-24 outages / cells / solves per coupled replay	37 / 864 / 31,968
Maximum counterfactual post-contingency loading (p.u.)	1.0000

capacity yield 12,296,675 variables through an exact separable knapsack (Table V); event energy is 0.194/0 MWh with zero residual and 29.486 MWh slack. Experiment 22 checks the typed invariant over 1,056 event slots and all finite RTS-24 outages: raw secure value is $0.004293 per interval without reoptimization. The identical indexed vector gives $0.006352 at 1.479689× fixed-nameplate and $9.213788 at 2146.204351× capacity-proportional scale; all replays pass 37 N–1 checks. Experiment 21 reports scale factors, not a second trajectory.

Experiment 25 independently validates the declaration boundary: 71,128/71,144 jobs match execution, requested-nameplate coverage is 1.000, and no window is infeasible under the committed rule. Its 689-job stress cohort uses 0.001 MW per region and a 0.004-MWh floor; 43.75% of capacity slots bind and minimum slack exceeds -10^{-19} MWh. This is separate from Experiment 19 (Table V).

Experiment 24 freezes two profiles and evaluates 37 finite RTS-24 outages over 54 days and 8 event slots (864 cells; 31,968 solves). Every cell succeeds with maximum loading 1.000000 p.u.; the outage count and bound are in Table V.

VI. Conclusion

This paper couples a provenance-bound workload verifier to signed nodal settlement. The validation-fitted total-plus-CVaR risk profile reaches nRMSE/false credit/F_1 = $0.314/1.809/0.470$ MWh/day on the complete-ledger mechanism-isolation panel, while the slot-60 gate replay uses only committed declarations and is reported separately (Table IV). The submitted-job witness spans 12,296,675 variables, yields 0.194 MWh net reduction with zero rebound, and maps to a zero-residual RTS-24 N–1 certificate at the raw, fixed-nameplate, and capacity-proportional scales. A 689-job stress cohort binds regional capacity at 0.001 MW while meeting the 0.004-MWh floor. Fixed facility demand is kept outside the conversion factor, so raw q_{99} is an auditable activation endpoint; the flexible-only capacity factor remains a planning diagnostic. Deployment still requires signed co-located telemetry and topology access, and causal utility validation remains external.

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